ONE COMMON SMOOTH NEIGHBOURHOOD FOR AN ADIC VERSAL FORMAL MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. The final source sentence reverses the ring arrow: the asserted smooth map is B_n to A_n, as in all preceding definitions, not A_n to B_n.

Editorial note. The source calls the associated graded rings Rees algebras; the displayed direct sums of successive quotients are the actual objects used.

Editorial note. The affine reduction sentence saying V-prime_n to W-prime_n is smooth means at the selected point before shrinking; the following sentence explicitly states the need for one common neighbourhood.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Locally Noetherian formal algebraic spaces and adic maps, Artin versality via square-zero lifting, algebraization of etale maps of affine completions, smoothness descent, finite-module support and the graded flatness criterion.

Source reference. Original section 07XZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10094–10094. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 07XF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10075–10075. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0CXI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16208–16208. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 045G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5435–5435. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 06EG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8215–8215. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Date: stacks-project:artin.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 07WL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10046–10046. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 07XG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10076–10076. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 06HM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8326–8326. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 07WK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10045–10045. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 02HX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3280–3280. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0GH8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20573–20573. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 07WQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10050–10050. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0GHM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20587–20587. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0AQ2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13495–13495. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 03XZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5166–5166. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains

the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0AN2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13425–13425. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0AR4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13532–13532. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0AS8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13571–13571. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00TF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1073–1073. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Lemma 0.1. *Let S be a locally Noetherian scheme. Let V be a scheme locally of finite type over S. Let $Z \subset V$ be closed. Let W be a locally Noetherian formal algebraic space over S such that W_{red} is locally of finite type over S. Let $g : V_{/Z} \to W$ be an adic morphism of formal algebraic spaces over S. Let $v \in V$ be a closed point such that g is versal at v (as in Section 07XZ). Then after replacing V by an open neighbourhood of v the morphism g is smooth (see proof).*

Lemma 0.2. *Let S, $\mathcal{X}$, U, x, u_0 be as in Definition 07XF. Assume*

(1) *$\Delta : \mathcal{X} \to \mathcal{X} \times \mathcal{X}$ is representable by algebraic spaces,*
(2) *Δ is locally of finite type (for example if $\mathcal{X}$ is limit preserving), and*
(3) *$\mathcal{X}$ has (RS).*

Let V be a scheme locally of finite type over S and let y be an object of $\mathcal{X}$ over V. Form the 2-fibre product

$$\begin{array}{ccc} \mathcal{Z} & \longrightarrow & (Sch/U)_{fppf} \\ \downarrow & & \downarrow{\scriptstyle x} \\ (Sch/V)_{fppf} & \xrightarrow{y} & \mathcal{X} \end{array}$$

Let Z be the algebraic space representing $\mathcal{Z}$ and let $z_0 \in |Z|$ be a finite type point lying over u_0. If x is versal at u_0, then the morphism $Z \to V$ is smooth at z_0.

Proof. (The parenthetical remark in the statement holds by Lemma 0CXI.) Observe that Z exists by assumption (1) and Algebraic Stacks, Lemma 045G. By assumption (2) we see that $Z \to V \times_S U$ is locally of finite type. Choose a scheme W, a closed point $w_0 \in W$, and an étale morphism $W \to Z$ mapping w_0 to z_0, see Morphisms of Spaces, Definition 06EG. Then W is locally of finite type over S and w_0 is a finite type point of W. Let $l = \kappa(z_0)$. Denote $z_{l,0}$, $v_{l,0}$, $u_{l,0}$, and $x_{l,0}$ the

objects of $\mathcal{Z}$, $(Sch/V)_{fppf}$, $(Sch/U)_{fppf}$, and $\mathcal{X}$ over $\mathrm{Spec}(l)$ obtained by pullback to $\mathrm{Spec}(l) = w_0$. Consider

$$\begin{array}{ccccc} \mathcal{F}_{(Sch/W)_{fppf},l,w_0} & \longrightarrow & \mathcal{F}_{\mathcal{Z},l,z_{l,0}} & \longrightarrow & \mathcal{F}_{(Sch/U)_{fppf},l,u_{l,0}} \\ & & \downarrow & & \downarrow \\ & & \mathcal{F}_{(Sch/V)_{fppf},l,v_{l,0}} & \longrightarrow & \mathcal{F}_{\mathcal{X},l,x_{l,0}} \end{array}$$

By Lemma 07WL the square is a fibre product of predeformation categories. By Lemma 07XG we see that the right vertical arrow is smooth. By Formal Deformation Theory, Lemma 06HM the left vertical arrow is smooth. By Lemma 07WK we see that the left horizontal arrow is smooth. We conclude that the map

$$\mathcal{F}_{(Sch/W)_{fppf},l,w_0} \to \mathcal{F}_{(Sch/V)_{fppf},l,v_{l,0}}$$

is smooth by Formal Deformation Theory, Lemma 06HM. Thus we conclude that $W \to V$ is smooth at w_0 by More on Morphisms, Lemma 02HX. This exactly means that $Z \to V$ is smooth at z_0 and the proof is complete. □

Lemma 0.3. *In Situation 0GH8 the functor F satisfies the Rim-Schlessinger condition (RS).*

Proof. Recall that the condition only involves the evaluation $F(V)$ of the functor F on schemes V over S which are spectra of Artinian local rings and the restriction maps $F(V_2) \to F(V_1)$ for morphisms $V_1 \to V_2$ of schemes over S which are spectra of Artinian local rings. Thus let V/S be the spetruim of an Artinian local ring. If $\xi = (Z, u', \hat{x}) \in F(V)$ then either $Z = \emptyset$ or $Z = V$ (set theoretically). In the first case we see that $\hat{x}$ is a morphism from the empty formal algebraic space into W. In the second case we see that u' is a morphism from the empty scheme into X' and we see that $\hat{x} : V \to W$ is a morphism into W. We conclude that

$$F(V) = U'(V) \amalg W(V)$$

and moreover for $V_1 \to V_2$ as above the induced map $F(V_2) \to F(V_1)$ is compatible with this decomposition. Hence it suffices to prove that both U' and W satisfy the Rim-Schlessinger condition. For U' this follows from Lemma 07WQ. To see that it is true for W, we write $W = \mathrm{colim}\, W_n$ as in Formal Spaces, Lemma 0GHM. Say $V = \mathrm{Spec}(A)$ with $(A, \mathfrak{m})$ an Artinian local ring. Pick $n \geq 1$ such that $\mathfrak{m}^n = 0$. Then we have $W(V) = W_n(V)$. Hence we see that the Rim-Schlessinger condition for W follows from the Rim-Schlessinger condition for W_n for all n (which in turn follows from Lemma 07WQ). □

Proof. Since g is adic it is representable by algebraic spaces (Formal Spaces, Section 0AQ2). Thus by saying g is smooth we mean that g should be smooth in the sense of Bootstrap, Definition 03XZ.

Write $W = \mathrm{colim}\, W_n$ as in Formal Spaces, Lemma 0GHM. Set $V_n = V_{/Z} \times_{\hat{x},W} W_n$. Then V_n is a closed subscheme with underlying set Z. Smoothness of $V \to W$ is equivalent to the smoothness of all the morphisms $V_n \to W_n$ (this holds because any morphism $T \to W$ with T a quasi-compact scheme factors through W_n for some n). We know that the morphism $V_n \to W_n$ is smooth at v by Lemma 0.2[1]. Of course this means that given any n we can shrink V such that $V_n \to W_n$ is smooth. The problem is to find an open which works for all n at the same time.

[1]The lemma applies since the diagonal of W is representable by algebraic spaces and locally of finite type, see Formal Spaces, Lemma 0AN2 and we have seen that W has (RS) in the proof of Lemma 0.3.

The question is local on V, hence we may assume $S = \mathrm{Spec}(R)$ and $V = \mathrm{Spec}(A)$ are affine.

In this paragraph we reduce to the case where W is an affine formal algebraic space. Choose an affine formal scheme W' and an étale morphism $W' \to W$ such that the image of v in W_{red} is in the image of $W'_{red} \to W_{red}$. Then $V_{/Z} \times_{g,W} W' \to V_{/Z}$ is an adic étale morphism of formal algebraic spaces over S and $V_{/Z} \times_{g,W} W'$ is an affine formal algebraic space. By Algebraization of Formal Spaces, Lemma 0AR4 there exists an étale morphism $\varphi : V' \to V$ of affine schemes such that the completion of V' along $Z' = \varphi^{-1}(Z)$ is isomorphic to $V_{/Z} \times_{g,W} W'$ over $V_{/Z}$. Observe that v is the image of some $v' \in V'$. Since smoothness is preserved under base change we see that $V'_n \to W'_n$ is smooth for all n. In the next paragraph we show that after replacing V' by an open neighbourhood of v' the morphisms $V'_n \to W'_n$ are smooth for all n. Then, after we replace V by the open image of $V' \to V$, we obtain that $V_n \to W_n$ is smooth by étale descent of smoothness. Some details omitted.

Assume $S = \mathrm{Spec}(R)$, $V = \mathrm{Spec}(A)$, $Z = V(I)$, and $W = \mathrm{Spf}(B)$. Let v correspond to the maximal ideal $I \subset \mathfrak{m} \subset A$. We are given an adic continuous R-algebra homomorphism

$$B \longrightarrow A^\wedge$$

Let $\mathfrak{b} \subset B$ be the ideal of topologically nilpotent elements (this is the maximal ideal of definition of the Noetherian adic topological ring B). Observe that $\mathfrak{b}A^\wedge$ and $IA^\wedge$ are both ideals of definition of the Noetherian adic ring $A^\wedge$. Also, $\mathfrak{m}A^\wedge$ is a maximal ideal of $A^\wedge$ containing both $\mathfrak{b}A^\wedge$ and $IA^\wedge$. We are given that

$$B_n = B/\mathfrak{b}^n \to A^\wedge/\mathfrak{b}^n A^\wedge = A_n$$

is smooth at $\mathfrak{m}$ for all n. By the discussion above we may and do assume that $B_1 \to A_1$ is a smooth ring map. Denote $\mathfrak{m}_1 \subset A_1$ the maximal ideal corresponding to $\mathfrak{m}$. Since smoothness implies flatness, we see that: for all $n \geq 1$ the map

$$\mathfrak{b}^n/\mathfrak{b}^{n+1} \otimes_{B_1} (A_1)_{\mathfrak{m}_1} \longrightarrow \left(\mathfrak{b}^n A^\wedge/\mathfrak{b}^{n+1}A^\wedge\right)_{\mathfrak{m}_1}$$

is an isomorphism (see Algebra, Lemma 0AS8). Consider the Rees algebra

$$B' = \bigoplus\nolimits_{n\geq 0} \mathfrak{b}^n/\mathfrak{b}^{n+1}$$

which is a finite type graded algebra over the Noetherian ring B_1 and the Rees algebra

$$A' = \bigoplus\nolimits_{n\geq 0} \mathfrak{b}^n A^\wedge/\mathfrak{b}^{n+1}A^\wedge$$

which is a a finite type graded algebra over the Noetherian ring A_1. Consider the homomorphism of graded A_1-algebras

$$\Psi : B' \otimes_{B_1} A_1 \longrightarrow A'$$

By the above this map is an isomorphism after localizing at the maximal ideal $\mathfrak{m}_1$ of A_1. Hence $\mathrm{Ker}(\Psi)$, resp. $\mathrm{Coker}(\Psi)$ is a finite module over $B' \otimes_{B_1} A_1$, resp. A' whose localization at $\mathfrak{m}_1$ is zero. It follows that after replacing A_1 (and correspondingly A) by a principal localization we may assume Ψ is an isomorphism. (This is the key step of the proof.) Then working backwards we see that $B_n \to A_n$ is flat, see Algebra, Lemma 0AS8. Hence $A_n \to B_n$ is smooth (as a flat ring map with smooth fibres, see Algebra, Lemma 00TF) and the proof is complete. □

References